

1. Class relationships

- 1.1 Association-- Associations are typically named using a verb or verb phrase which reflects the real world problem domain, possibly with multiplicities.

1.2 Aggregation

A special type of association.

- It represents a "part of" relationship.
- Class2 is part of Class1.
- Many instances (denoted by the *) of Class2 can be associated with Class1.
- Objects of Class1 and Class2 have separate lifetimes

1.3 Composition

- A special type of aggregation where parts are destroyed when the whole is destroyed.
- Objects of Class2 live and die with Class1.
- Class2 cannot stand by itself.

1.4 Generalization (Inheritance)

A generalization is a taxonomic relationship between a more general classifier and a more specific classifier:

- Represents an "is-a" relationship.
- An abstract class name is shown in italics.
- SubClass1 and SubClass2 are specializations of SuperClass